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| **Fiscal Year:** | FY 2004 | **Task Last Updated:** | FY 03/23/2006 |
| **PI Name:** | Day, Delbert E. Ph.D. | | |
| **Project Title:** | Kinetics of Nucleation and Crystal Growth in Glass Forming Melts in Microgravity | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | MATERIALS SCIENCE | | |
| **Program/Discipline--Element/Subdiscipline:** | MATERIALS SCIENCE--Glasses and ceramics | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | day@umr.edu or csray@umr.edu | **Fax:** | FY 573-341-2071 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 573-341-4354 |
| **Organization Name:** | University of Missouri, Rolla | | |
| **PI Address 1:** | Materials Research Center | | |
| **PI Address 2:** | 109 Straumanis Hall | | |
| **PI Web Page:** | http://science.nasa.gov/ ; http://www.spacedaily.com/ http://web.umr.edu/ | | |
| **City:** | Rolla | **State:** | MO |
| **Zip Code:** | 65409-1170 | **Congressional District:** | 8 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 00-HEDS-02 |
| **Start Date:** | 02/01/1998 | **End Date:** | 01/31/2004 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA MSFC |
| **Contact Monitor:** | Ethridge, Ed | **Contact Phone:** | 256 544-7677 |
| **Contact Email:** | ed.ethridge@msfc.nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | Day, Delbert | | |
| **COI Name (Institution):** | Ray, C. S. ( University of Missouri at Rolla ) | | |
| **Grant/Contract No.:** | NAG8-1465 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

The investigative approach of this work will be to conduct ground-based as well as flight experiments with an aim to obtaining data for the kinetics of nucleation and crystal growth, and values for various properties of a lithium disilicate glass. The ground-based work is divided into two parts. The first part is concerned with studying the glasses prepared on Earth and the second part deals with similar measurements of the glasses prepared in space. In general, the ground-based work includes:
(1) melting glasses for ground-based and flight experiments;

(2) measuring the nucleation (I) and crystal growth (U) rates as a function of temperature (T) using the conventional method of two stage heat treatment followed by the microscopic measurement of the number and size of crystals for glasses prepared in space and on Earth;

(3) measuring the concentration of quenched-in nuclei (Nq), I and U for the glasses prepared in space and on earth using

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| **Task Description:** | a differential thermal analysis (DTA) method developed under the present contract;<br><br>(4) measuring selected properties such as chemical homogeneity, density, refractive index, transmission in the UV, visible, and IR, specific heat capacity, glass transition and crystallization temperatures for glasses prepared in space and on Earth;<br><br>(5) investigating the effect of isostatic pressure and shear stress on the nucleation and crystallization in glasses;<br><br>(6) analysis of glass structure by IR, Raman, and X-ray photoelectron spectroscopy and neutron diffraction for glasses prepared in space and on Earth;<br><br>(7) computer simulation of the effect of compositional fluctuation induced by gravity-driven convection in a melt on the size and number density of the evolving clusters;<br><br>(8) preparation of precursor samples (glass encapsulated in doubly sealed cylindrical platinum containers) for use in flight experiments.<br><br>The measurements in steps 2 to 3 above will yield the following results.<br><br>(a) Nq for glasses prepared in space and on Earth;<br><br>(b) I and U as a function of T for the glasses prepared in space and on Earth, but nucleated and crystallized on earth;<br><br>(c) I and U as a function of T for the glasses prepared in space and on earth, but nucleated and crystallized in space.<br><br>A comparison of the results in (a) to (c) should identify whether the nucleation rate, crystal growth rate, or the concentration of quenched-in nuclei for this lithium disilicate glass is different in micro-g and at 1g.<br><br>The flight experiments will include:<br><br>(1) glass melting at 14000 C, in sealed platinum capsules, about 2.5 cm long and 1.0 cm in diameter;<br><br>(2) heat treatment (for measuring I) at three selected temperatures, 440, 455, and 4700 °C, for three different times (1, 3, and 5 h) at each temperature followed by a heat treatment for crystal growth at a fixed temperature (~ 6000 °C) for a fixed time (20 min), for glasses prepared in space and on Earth; and<br><br>(3) heat treatment (for measuring U) at three selected temperatures, 590, 610, and 6300 °C, for three different times (10, 20, and 30 min) at each temperature for glasses prepared in space and on Earth.<br><br>Some of the glasses prepared in space will be returned to earth in their as-quenched condition (i.e., without receiving any heat treatment for nucleation or crystal growth in space) for structural evaluation, measuring specific properties, and determining the nucleation and crystal growth rates at 1 g. A glass containing a colored dot on its surface will be remelted in space for determining the extent of fluid flow in the high temperature melts in microgravity.<br><br>High temperature melts in microgravity are reported to form glass more easily and to be more chemically homogeneous and more resistant to crystallization than identical glasses prepared at 1 g on Earth. No explanation currently exists for these intriguing and potentially useful results. The primary objective of this research is to investigate these fundamental issues, namely, why melts in microgravity form glass more easily and why the chemical homogeneity of these glasses is better than that of the glasses made on Earth (1 g). Since glasses are traditionally prepared by cooling a melt, the glass formation and properties of a glass, particularly, its chemical homogeneity and crystallization tendency, are controlled by the kinetics of nucleation and crystal growth. The best way to verify and understand these important scientific questions is to measure the rates of nucleation and crystal growth for glasses prepared in micro-g and to compare these results with values measured on the same glass prepared at 1 g. This comparison will be made in the present investigation using a lithium disilicate glass, since its crystallization properties at 1g are well known.<br><br>A second objective of this research will be to perform a complete characterization of the space and Earth-melted glasses through measurements of their density, refractive index, transmission in the UV, visible and infrared (IR), and glass transition and crystallization temperatures. Any difference in the structure of the space- and Earth-melted glasses will also be investigated by x-ray photoelectron spectroscopy (XPS), neutron diffraction, and Raman and IR spectral analysis.<br><br>The reported results on the improvement in the quality of space-melted glasses compared to glasses prepared on earth, have not been given due attention largely because existing theories do not explain these results. The present research is designed to answer the critical question, namely, why the quality, the chemical homogeneity and resistance to crystallization in particular, is improved for a space-melted glass. The results of this research should lead to a better understanding of the fundamental mechanisms of glass formation, nucleation, and crystallization of high temperature oxide melts, which can improve present-day glass processing technology on Earth. An improved processing technology could yield glasses with a quality superior to what is presently attainable and could help advance applications in communication, optical switching and devices where glasses are used as the key functional material. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | No progress report this reporting period. |
| **Bibliography Type:** | Description: (Last Updated: 02/27/2004) |